Zikang Xu

Date of Birth Jan., 1997
Expected Graduate Date July, 2025

Education

School of BME, **University of Science and Technology of China** — Suzhou, China
Ph.D. candidate — Sept.2021 until now
Supervisor：Prof. S. Kevin Zhou (Fellow of IEEE, AIMBE, NAI)

School of BME, **Southeast University** — Nanjing, China
M. Eng. — Sept. 2018 - July 2021
Supervisor：Prof. Zhongze Gu (Fellow of AIMBE, RSC)
Thesis: Microscopic Image Reconstruction and Enhancement Algorithm Based on Deep Learning

School of BME, **Southeast University** — Nanjing, China
Chien-Shiung Wu College (honor college), **Southeast University** — Nanjing, China
B. Eng. — Sept. 2014 - July 2018
Thesis: Machine Learning Based Emotion Recognition of EEG Signals

Research Interests

During my Ph.D. studies, I specialized in **Algorithm Fairness in Medical Image Analysis**, and **Landmark Detection** with my research published (as first and second co-author) in **1** top-tier journal and **3** conferences, including NPJ Dig. Med., MICCAI, and MIDL. I am also interested in **Medical Segment Anything Models**, **Surgical Instrument Segmentation**, and **Biomedical Image Analysis**.

Publications

Algorithm Fairness in Medical Image Analysis

[MICCAI-2021] FairAdaBN: Mitigating unfairness with adaptive batch normalization and its application to dermatological disease classification
Zikang Xu, Shang Zhao, Quan Quan, Qingsong Yao, S. Kevin Zhou

[NPJ Dig. Med.] Addressing Fairness Issues in Deep Learning-Based Medical Image Analysis: A Systematic Review
Zikang Xu, Jun Li, Qingsong Yao, Han Li, Mingyue Zhao, S. Kevin Zhou

[NeurIPS 2024 in submission] FairMedFM: Fairness Benchmarking for Medical Imaging Foundation Models
Ruinan Jin*, **Zikang Xu***, Yuan Zhong*, Qingsong Yao, Xiaoxiao Li, S. Kevin Zhou, Qi Dou

[In Preparation] APPLE: Adversarial Protected-attribute-aware Perturbations on Latent Embedding
Zikang Xu, Fenghe Tang, Quan Quan, Qingsong Yao, S. Kevin Zhou

[ArXiv] Inspecting Model Fairness in Ultrasound Segmentation Tasks
Zikang Xu, Fenghe Tang, Quan Quan, Jianrui Ding, Chunping Ning, S. Kevin Zhou

Catheter Landmark Detection

[MICCAI-2024] SIX-Net: Spatial-context Information miX-up for Electrode Landmark Detection
Xinyi Wang, **Zikang Xu**, Qingsong Yao, Heqin Zhu, Yiyong Sun, S. Kevin Zhou

[MIDL-2024] OFELIA: Optical Flow-based Electrode LocalIzAtion
Xinyi Wang, **Zikang Xu**, Qingsong Yao, Yiyong Sun, S. Kevin Zhou

Others

[MIDL-2024, Oral] Slide-SAM: Medical SAM Meets Sliding Window
Quan Quan, Fenghe Tang, **Zikang Xu**, Heqin Zhu, S. Kevin Zhou

[MedIA, Major Revision] Qiyuan Wang, Shang Zhao, **Zikang Xu**, S. Kevin Zhou

[Biosensors and Bioelectronics] A storm in a teacup–A biomimetic lung micro-physiological system in conjunction with a deep-learning algorithm to monitor lung pathological and inflammatory reactions
Zaozao Chen, Jie Huang, Jing Zhang, **Zikang Xu**, Qiwei Li, Jun Ouyang, Yuchuan Yan, Shiqi Sun, Huan Ye, Fei Wang, Jianfeng Zhu, Zhangyan Wang, Jie Chao, Yuepu Pu, Zhongze Gu

Research Experience

Fuwai Hospital, Chinese Academy of Medical Sciences , Research intern. Beijing, China
Radiography Department June 2023 - Oct. 2023

Journal and Conference Reviewer

IEEE Trans. Med. Imaging, MICCAI 2024

Awards

- 2016 National Biomedical Engineering Innovation Design Competition for College Students
- 2021 Suzhou Industrial Park Scholarship
- 2021,2023 Academic First Class Scholarship in USTC
- 2022 Academic Second Class Scholarship in USTC

Talents

- IELTS 7.0 (Listening: 7.5, Reading: 7.0, Writing: 7.0, Speaking 6.0)
- Table-tennis, Badminton, Photography
- Python, MATLAB, PyTorch, LaTeX

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